TELESCOPIC MAST

Technical Description and Operating Instructions

ХЖ4.115.120 ТО

Translated copy — machine-assisted translation from the original Russian-language manual. Original designations, part numbers, and technical-specification codes (e.g. «ХЖ4.115.120», «ХЖ0.411.000 ТУ») are retained unchanged.

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1. Introduction

This technical description and operating instructions manual is intended for the study of telescopic masts ХЖ4.115.120, ХЖ4.115.120-01, ХЖ4.115.120-02, ХЖ4.115.120-03, ХЖ4.115.120-04, ХЖ4.115.120-05, ХЖ4.115.120-06, ХЖ4.115.120-07, ХЖ4.115.120-08, and ХЖ4.115.120-09.

In terms of design, masts ХЖ4.115.120-01, ХЖ4.115.120-02, ХЖ4.115.120-03, ХЖ4.115.120-04, ХЖ4.115.120-05, ХЖ4.115.120-07, ХЖ4.115.120-08, and ХЖ4.115.120-09 differ from mast ХЖ4.115.120 in the number of sections.

Masts ХЖ4.115.120-08 and ХЖ4.115.120-09 provide the greatest lifting height for the antenna-feeder device.

Masts ХЖ4.115.120-03, ХЖ4.115.120-04, ХЖ4.115.120-06, ХЖ4.115.120-07, ХЖ4.115.120-08, and ХЖ4.115.120-09 differ from mast ХЖ4.115.120-01 in the position of the limiting key on the second section, which determines the lifting height (see Table 1).

TECHNICAL DESCRIPTION

2. Telescopic Mast

2.1. Purpose

The telescopic mast is designed to raise antenna devices. The mast can be mounted on a vehicle chassis or body, and it can also be installed on the ground; for this purpose, the delivery set includes a base plate and first-tier guy wires.

2.2. Technical Data

The technical data of the telescopic masts are summarized in Table 1. The time required to deploy the mast with a crew of five is no more than 20 minutes.

Deployment (retraction) of the telescopic mast is only possible with an antenna device of at least 10 kg installed. The telescopic mast remains operable within a temperature range of minus 50°C to plus 50°C, at wind speeds of up to 30 m/s.

Table 1

Designation	Height of fully deployed mast, m, max	Mass of lifted load, kg, max	Height of mast folded (transport position), m	No. of sections	Force at winch handle, N, max — at start	Force at winch handle, N, max — at end	Force at winch handle, N, max — when retracting	Max. horizontal load on mast top, N (kgf)	Mast mass, kg	Technical specification for mast
ХЖ4.115.120	10.5	35	2.48	6	100	250	100	520 (52)	54.5	ХЖ0.411.000 ТУ
ХЖ4.115.120-01	12.1	35	2.6	7	100	250	100	520 (52)	60	ХЖ0.411.000 ТУ
ХЖ4.115.120-02	13.3	30	2.7	8	100	280	100	455 (45.5)	65.5	ХЖ0.411.000 ТУ
ХЖ4.115.120-03	7.2	35	2.6	7	100	550	100	520 (52)	60	ХЖ0.411.000 ТУ
ХЖ4.115.120-04	9.5	35	2.6	7	100	550	100	520 (52)	60	ХЖ0.411.000 ТУ
ХЖ4.115.120-05	7.2	35	2.28	4	100	250	100	520 (52)	45.5	ХЖ0.411.000 ТУ
ХЖ4.115.120-06	12.0	50	2.911	6	100	250	100	520 (52)	65	ХЖ0.411.000 ТУ
ХЖ4.115.120-07	8.3	70	2.745	4	100	250	100	1100 (110)	50.5	ХЖ0.411.000 ТУ
ХЖ4.115.120-08	13.9	30	3.0	7	100	280	100	455 (45.5)	70.5	ХЖ0.411.000 ТУ
ХЖ4.115.120-09	13.9	45	3.0	7	150	350	100	500 (50)	72.5	ХЖ0.411.000 ТУ

2.3. Composition

The telescopic mast (Fig. 1) consists of a fixed section, movable sections, pulleys, steel cables, and a winch.

2.4. Design and Principle of Operation

The movable sections of the telescopic mast are connected to one another and to the winch drum by a system of cables and pulleys. When the winch handle is turned clockwise, the cable winds onto the drum, which causes all of the mast's movable sections to extend simultaneously — that is, the telescopic mast deploys.

When the winch handle is turned counterclockwise, the cable unwinds from the drum, and the movable sections, under their own weight and the weight of the antenna device, retract one into another — that is, the telescopic mast retracts.

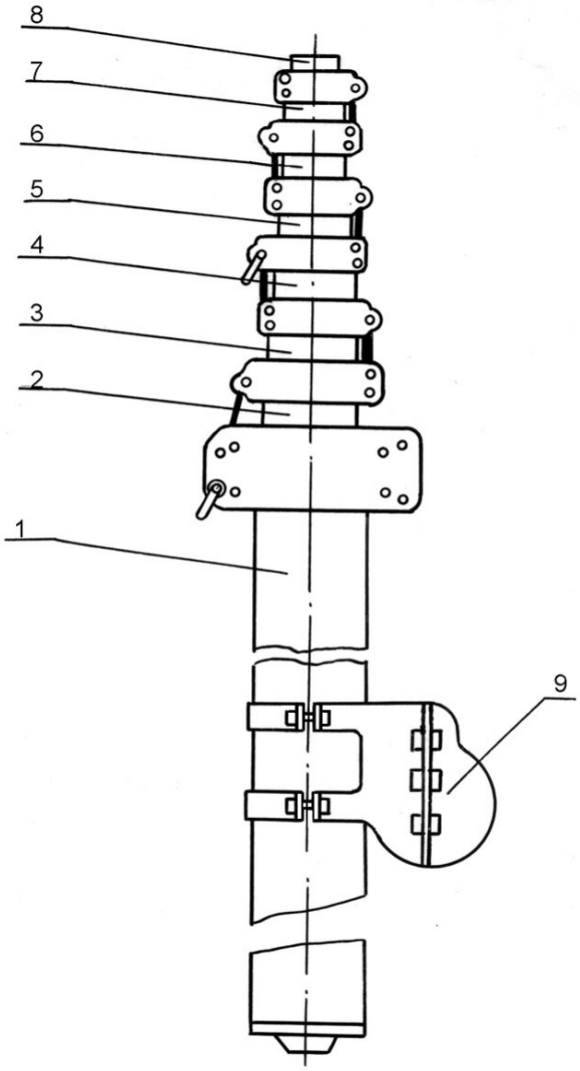


Fig. 1. Telescopic mast

1 – 1st section; 2 – 2nd section; 3 – 3rd section; 4 – 4th section; 5 – 5th section; 6 – 6th section; 7 – 7th section; 8 – 8th section; 9 – winch.

3. Design and Operation of Components

3.1. Fixed Section

The fixed first section (see Figs. 1, 2) is made from a duralumin tube.

3.1.1. Purpose and Design

Section 1 (the fixed section) houses the movable sections and provides the mounting for winch 9. A base 1 (Fig. 2) is secured in the opening of the lower end face. A clamp is fitted at the top of the fixed section (see Fig. 1); it consists of two halves cast from aluminum alloy and bolted together.

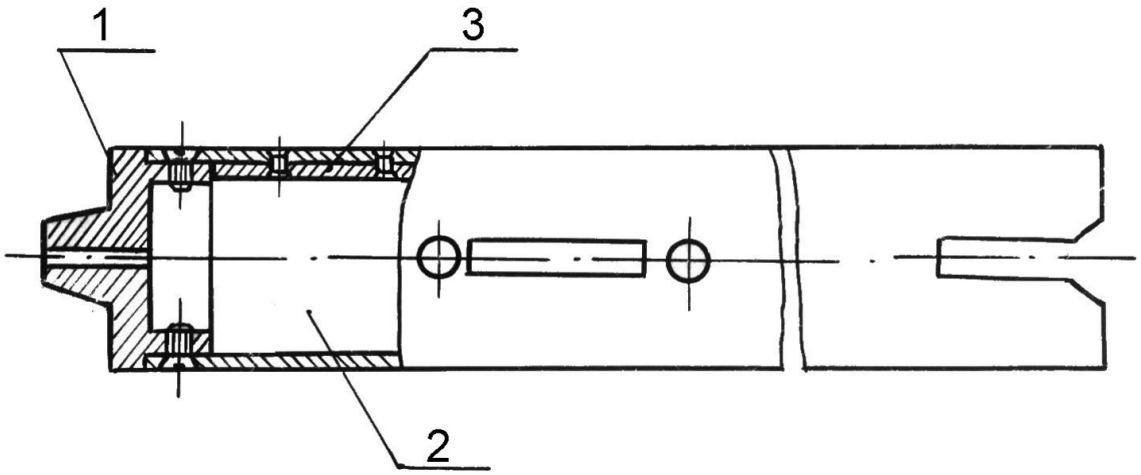


Fig. 2. First mast section

1 – base; 2 – tube; 3 – key.

3.2. Movable Sections

Movable sections 2–8 are made from tubes of aluminum alloy.

3.2.1. Purpose and Design

The design of the 2nd movable section differs from the others in that, on the side where the base is mounted, it has a cutout that allows the pulley mounted on base 2 (Fig. 3) to rotate.

To prevent the cable from slipping off pulley 5, a safety bracket 4 is mounted on the same axle as the pulley.

All of the other movable sections are identical in design and correspond to Fig. 4.

A band ring 3 is press-fitted and riveted onto the upper end of tube 5; it stiffens the section at the point where the clamp is fitted. An opening is made in band ring 3 and tube 5, leading into a slot that widens toward the upper end face. The opening allows the cable of the inserted section to pass through and be fitted onto the pulley (Figs. 5 and 8). The slot serves to hold the eccentric ring in place.

The ring's eccentricity creates a gap of 4.5–5 mm between the mast sections for the cable to pass through. The eccentric ring (Fig. 6), cast from aluminum alloy, has an insert made of polyamide resin that acts as a sealing element between the sections and, during deployment (retraction) of the mast, functions as a sliding bearing.

An eccentric base (Fig. 7), also cast from aluminum alloy, is fitted at the other end of the section.

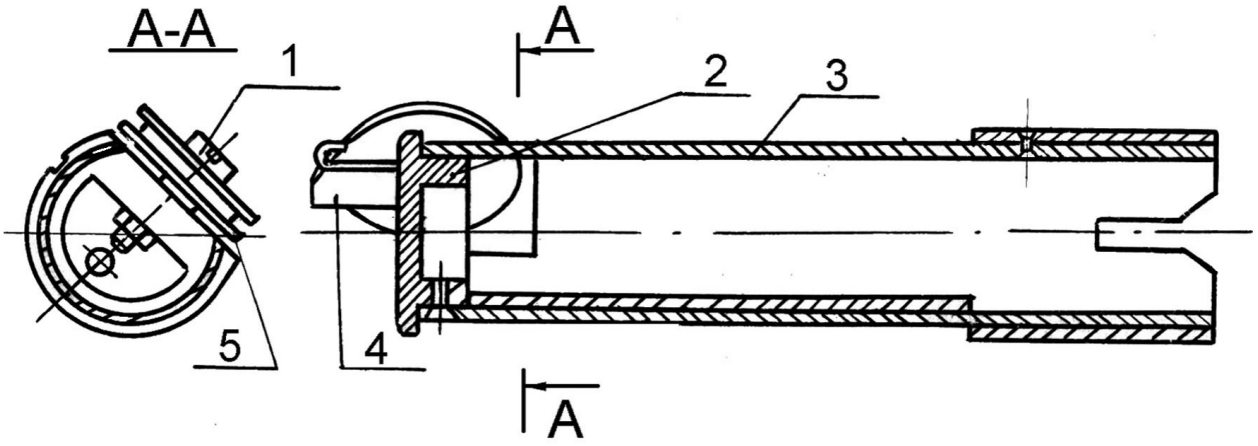


Fig. 3. Second (movable) mast section

1 – bolt; 2 – base; 3 – tube; 4 – bracket; 5 – pulley.

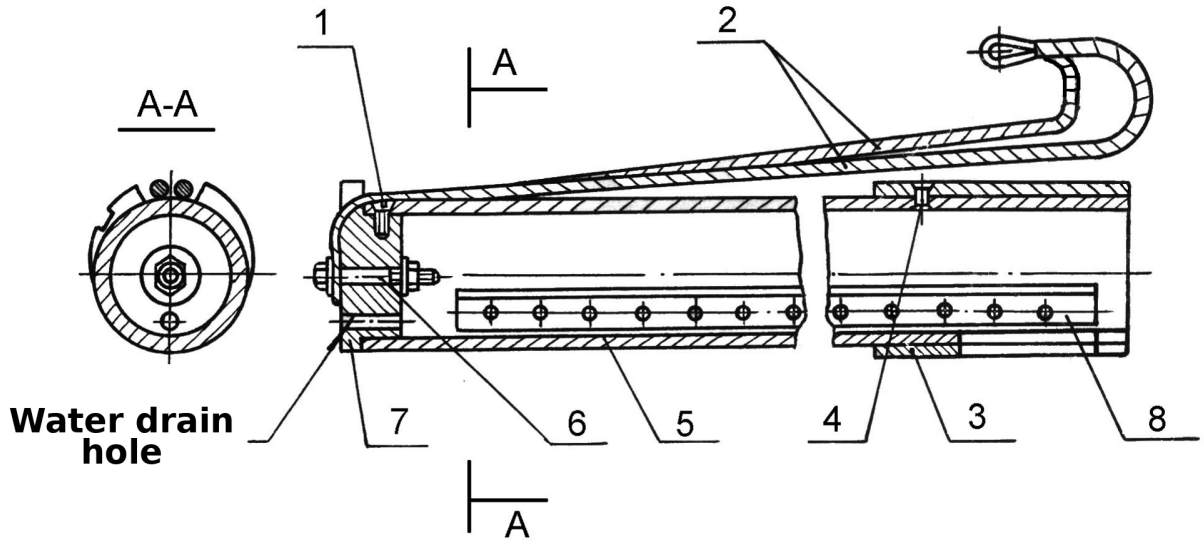


Fig. 4. Movable mast section

1 – screw; 2 – double cable; 3 – band ring; 4 – rivet; 5 – tube; 6 – bolt; 7 – base; 8 – key.

The eccentricity of the base and the ring is identical, and during assembly of the mast they are oriented to the same side; this ensures that the sections move in a straight line during deployment (retraction) of the mast. The base has a slot for the key, which is riveted inside each section except the topmost one. The key (Fig. 8) prevents one section from rotating relative to another.

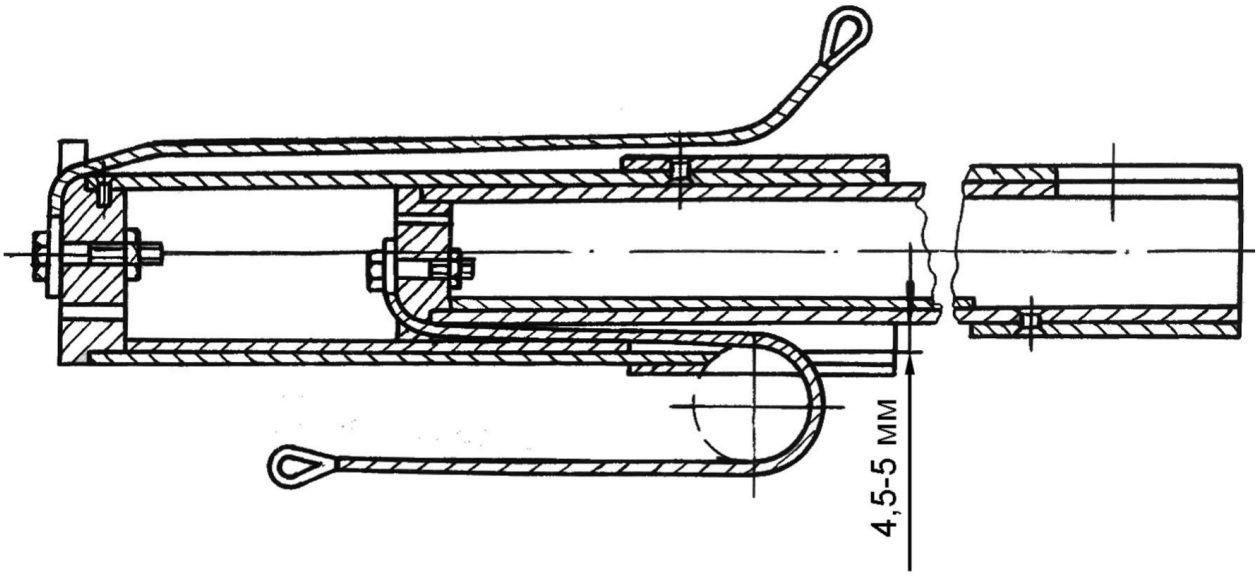


Fig. 5. Assembly of the mast's movable sections.

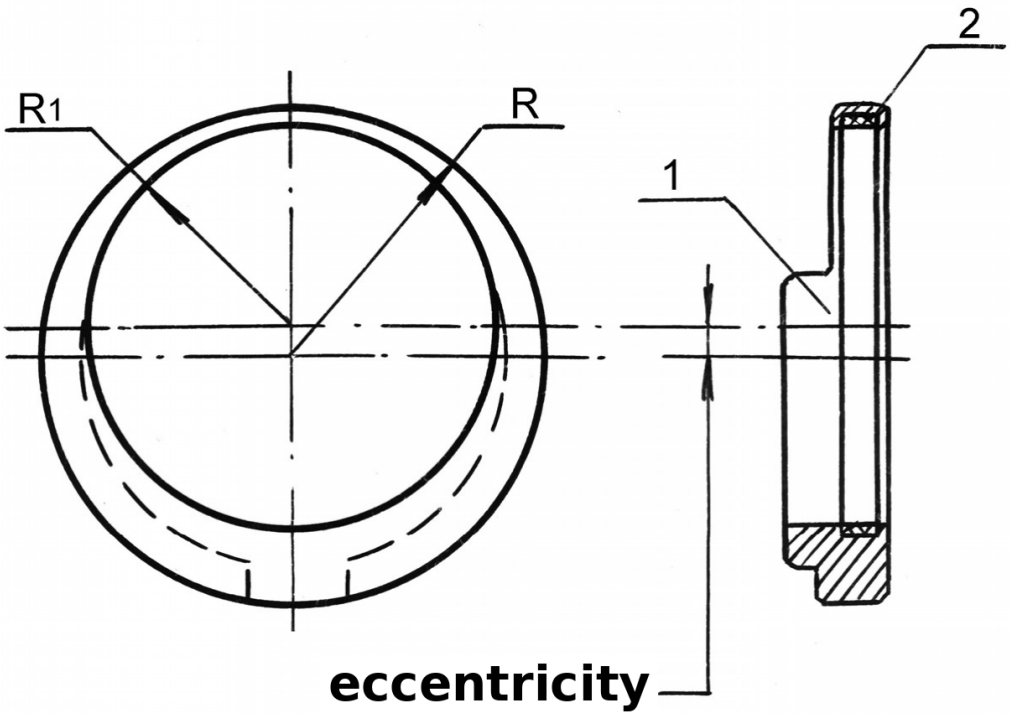


Fig. 6. Eccentric ring

1 – insert; 2 – sealing ring.

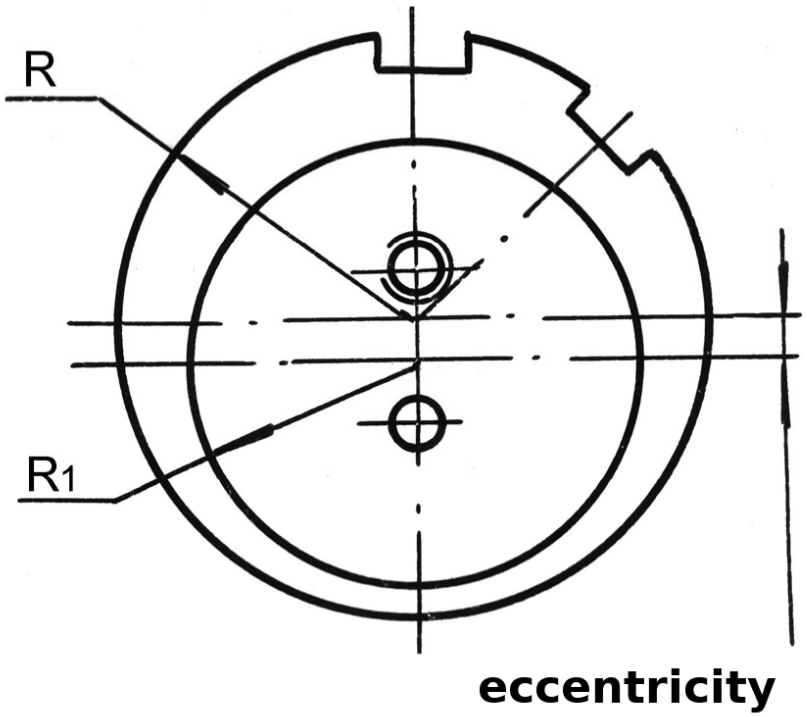


Fig. 7. Eccentric base.

3.3. Cables

A double steel cable, 4 mm in diameter, is fastened to the base of each movable section and laid into the slot (see Figs. 4, 5, 8).

The cables transmit force from the winch to the movable sections during deployment of the mast.

The two runs of the cable must be exactly equal in length to ensure an even load on each run.

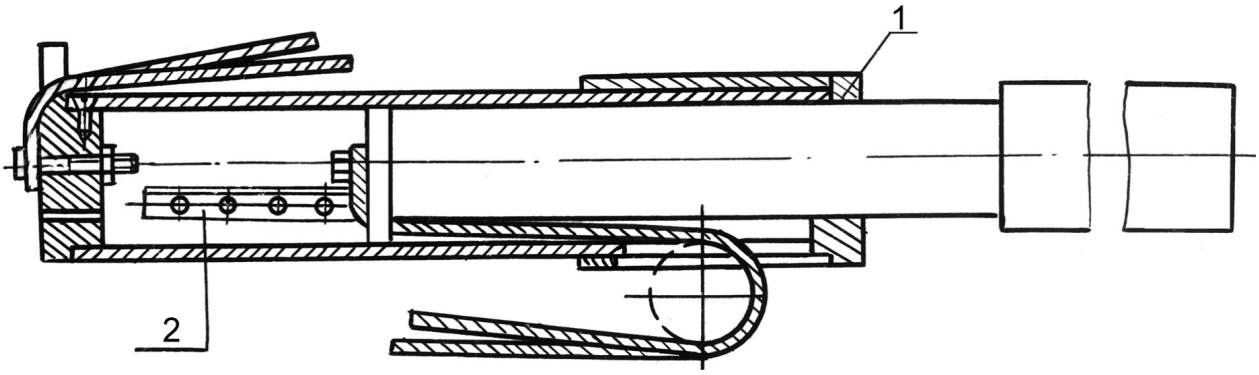


Fig. 8. Connection of the mast's movable sections

1 – ring; 2 – key.

3.4. Winch

Winch 9 (see Fig. 1) is a cylindrical gear reducer with horizontally arranged shafts, mounted in a housing that is closed by a cover. The housing and cover are cast from aluminum alloy.

3.4.1. Purpose and Design

The winch is designed to raise (lower) the mast's movable sections.

The winch mechanism, shown schematically in Fig. 9, consists of the following main parts:

lever 13;

handle 17;

gear 11, secured to the lever axle by key 12;

friction brake (see Fig. 10), which includes gear 6, bronze ring 5, ratchet wheel 4, conical bushing 3, gear 1, special nut 2, and brake axle 7. On the face toward special nut 2, gear 1 has a helical recess, and special nut 2 has a matching helical projection on the face toward gear 1. Gear 6 is secured to brake axle 7 by key 8. Bronze ring 5, ratchet wheel 4, conical bushing 3, and gear 1 are mounted loosely (free to rotate) on axle 7. Special nut 2 is secured to axle 7 by pin 9 so that the gap between the helical projection on its face and the helical recess on the face of gear 1 is within 0.2–0.4 mm;

the gear (see Fig. 9) of drum 1, onto which the cable winds;

and pin 19.

3.4.2. Operating Principle During Deployment

When the mast is deployed, handle 17 is turned clockwise (see Fig. 9). Gear 11 then transmits rotation to gear 16 and to special nut 2, which is rigidly fixed to axle 4 together with gear 16. Since the face of the special nut has a helical projection, and gear 5 has a matching helical recess, as soon as rotation begins the nut's projection presses against the recess in gear 5 and pushes gear 5 to the right.

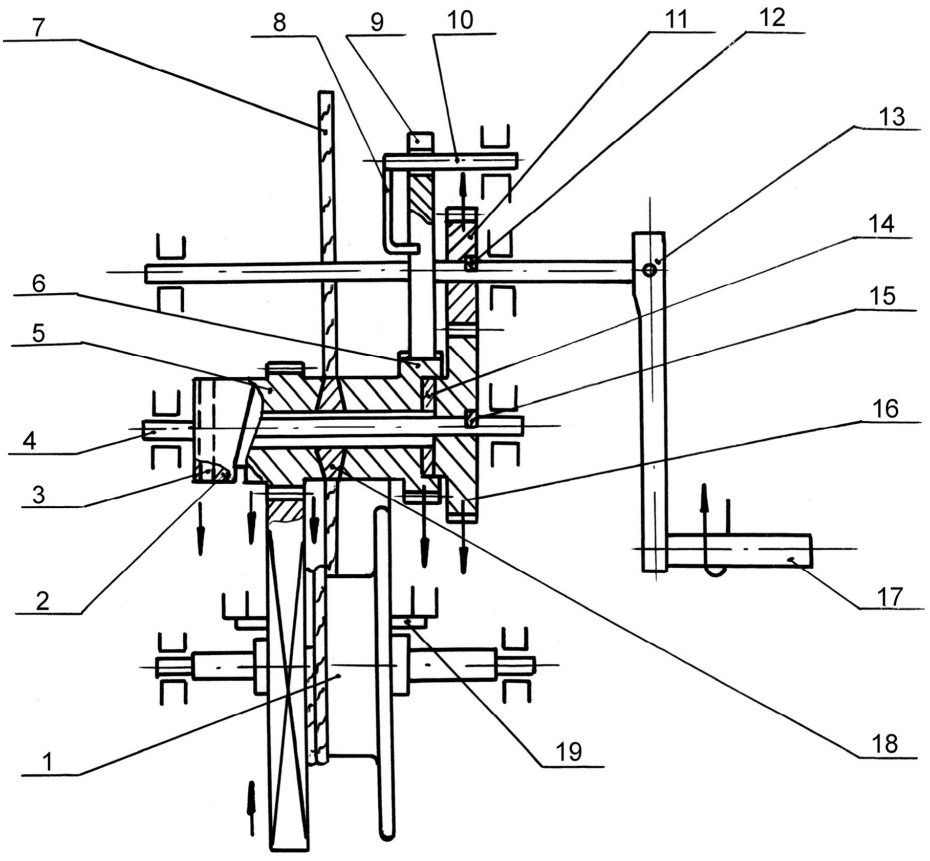


Fig. 9. Winch mechanism diagram

(arrows show the direction of motion during deployment of the mast)
1 – drum; 2 – special nut; 3 – pin; 4 – axle; 5 – special gear; 6 – ratchet wheel; 7 – cable; 8 – pawl spring; 9 – pawl; 10 – pawl axle; 11 – gear; 12 – key; 13 – lever; 14 – bronze ring; 15 – key; 16 – gear; 17 – handle; 18 – conical bushing; 19 – pin.

Gear 5 then frictionally engages conical bushing 18 with ratchet wheel 6. As a result, all the parts on the brake axle rotate in the same direction. Gear 5 transmits rotation to the gear of drum 1, to which one end of the cable is fixed, and the cable winds onto the drum. As the cable winds onto the drum, the mast's second section is raised; the second section raises the third, the third raises the fourth, and so on.

This results in the simultaneous raising of all the sections (Fig. 11). During deployment of the mast, pawl 9 (see Fig. 9) is continuously pressed by a spring against the teeth of ratchet wheel 6, which is why characteristic clicking sounds are heard in the winch mechanism.

If rotation of the handle is stopped while the mast is being deployed, the force applied to the winch lever and transmitted to gear 5 is replaced by a force made up of the weight of the raised part of the mast and the weight of the antenna device. This force acts on the teeth of drum gear 1 in the direction of arrow A (Fig. 12) and prevents gear 5 (see Fig. 9) from disengaging from the frictional coupling. Since ratchet wheel 6 is thereby locked, it prevents the entire friction brake (see Fig. 10) from turning, because pawl 9 (see Fig. 9) engages in a tooth gap of ratchet wheel 6 — so the winch mechanism will not rotate.

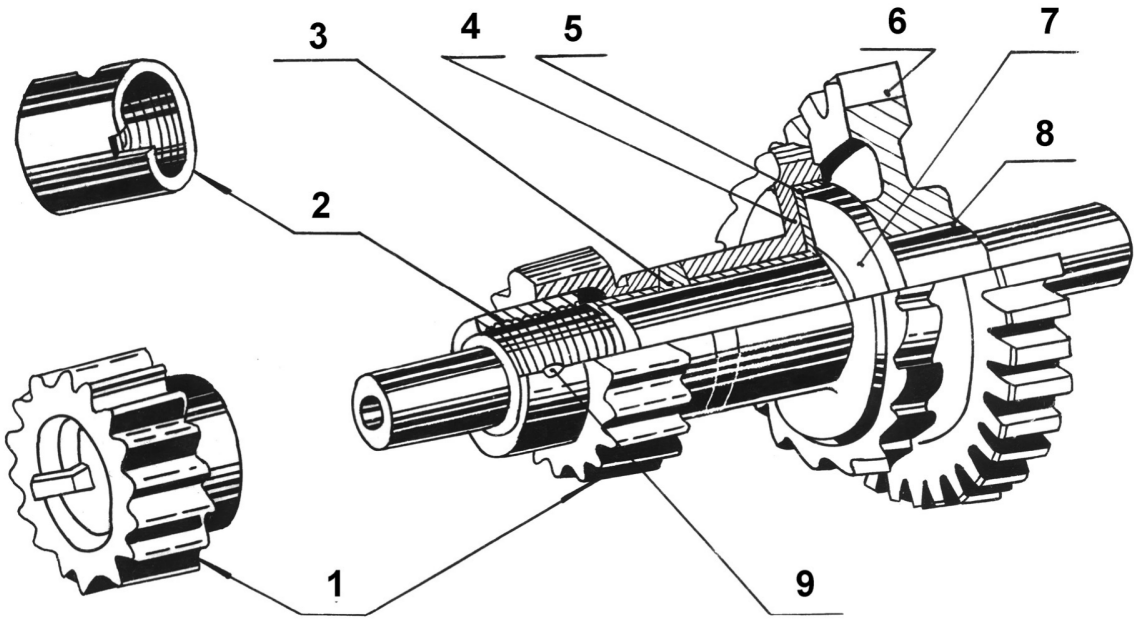


Fig. 10. Winch friction brake

1 – gear; 2 – special nut; 3 – conical bushing; 4 – ratchet wheel; 5 – bronze ring; 6 – gear; 7 – brake axle; 8 – key; 9 – pin.

When the mast reaches its maximum height, the limiting key fixed on the second section presses against the insert in the fixed section, making further clockwise rotation of handle 17 impossible.

3.4.3. Operating Principle During Retraction

When the mast is retracted, handle 17 (see Fig. 9) is turned counterclockwise. As a result, the entire mechanism rotates in the direction opposite to that shown by the arrows. The shoulder of the helical projection on special nut 2 approaches the edge of the helical recess on gear 5. A gap of 0.2–0.4 mm forms between special nut 2 and gear 5 (Fig. 13). Since gear 5 (see Fig. 9) is mounted loosely on axle 4, it shifts to the left along this axle and the frictional engagement is released.

Gear 5 then begins to turn the gear of drum 1. The cable unwinds from the drum. At the same time, under the weight of all the movable sections and the antenna device, the sections begin to retract one into another until their clamps meet. Once the clamps have

met, stop turning the handle.

Once the mast is fully retracted, further counterclockwise rotation of the handle becomes impossible, because pin 19, moving in the slot of the drum, comes up against the projections on the housing. This prevents the winch cable from becoming tangled.

If rotation of the handle is stopped while the mast is being retracted, force “A” (see Fig. 12), which acts on the cable in the same direction during both deployment and retraction, will cause the entire winch mechanism to turn in the reverse direction. Because the teeth of drum gear 1 (see Fig. 9) press on the teeth of gear 5, gear 5 is shifted to the right by the action of the helical projection and re-establishes the frictional coupling. Pawl 9, engaging in a tooth gap of ratchet wheel 6, prevents the entire friction brake from turning, and the winch mechanism will not rotate. As a result, the mast cannot retract on its own.

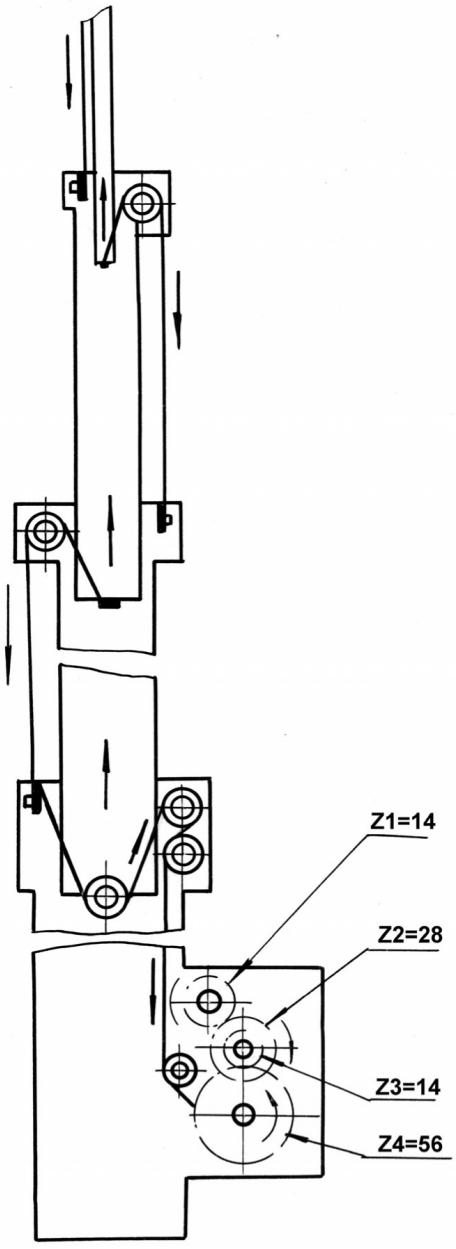


Fig. 11. Kinematic diagram of the telescopic mast.

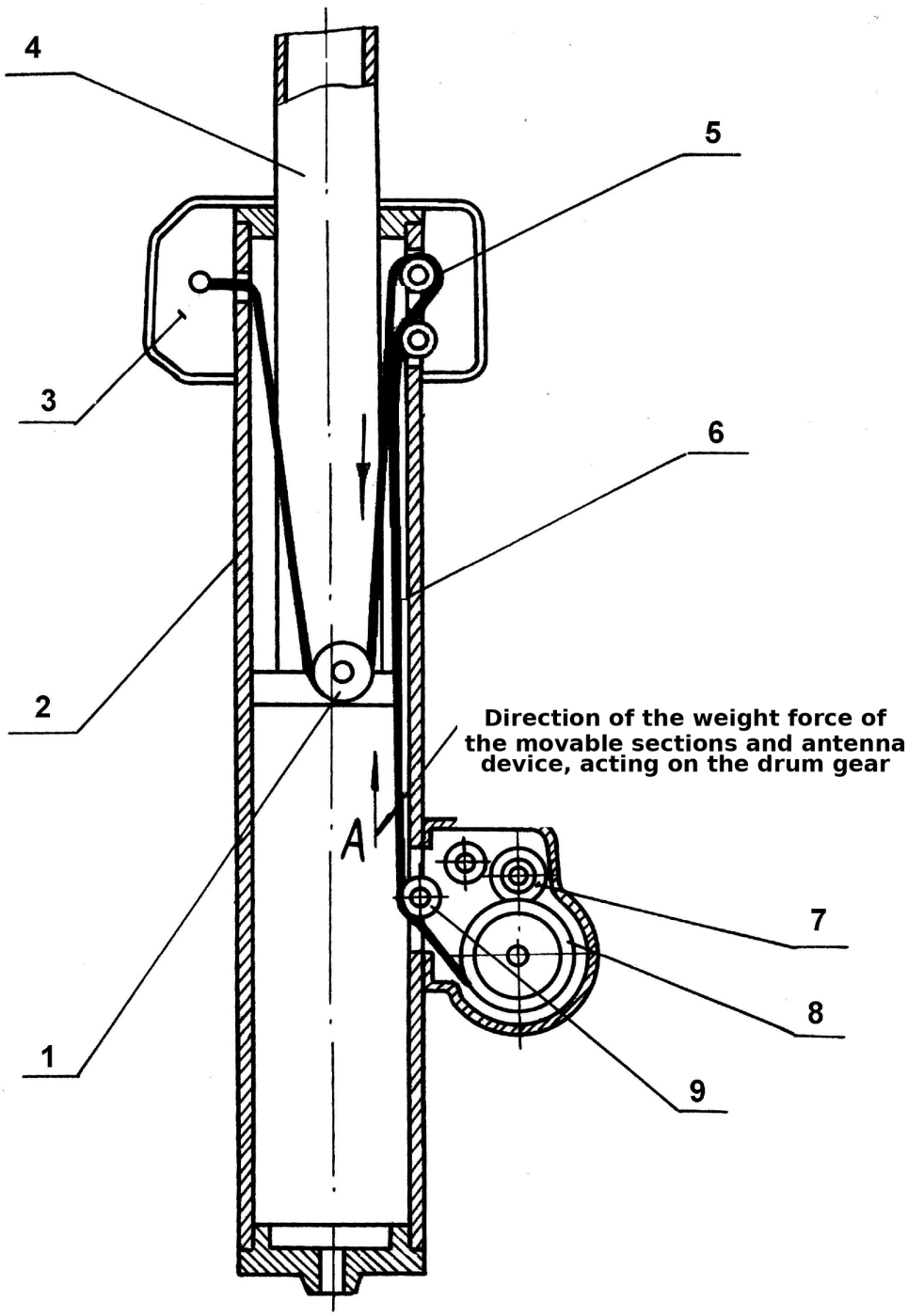


Fig. 12. Diagram of the connection of the second section's cable to the first section and the winch drum

1 – pulley; 2 – 1st section; 3 – clamp; 4 – 2nd section; 5 – pulley; 6 – cable; 7 – friction brake gear; 8 – drum with gear; 9 – cable guide pulley.

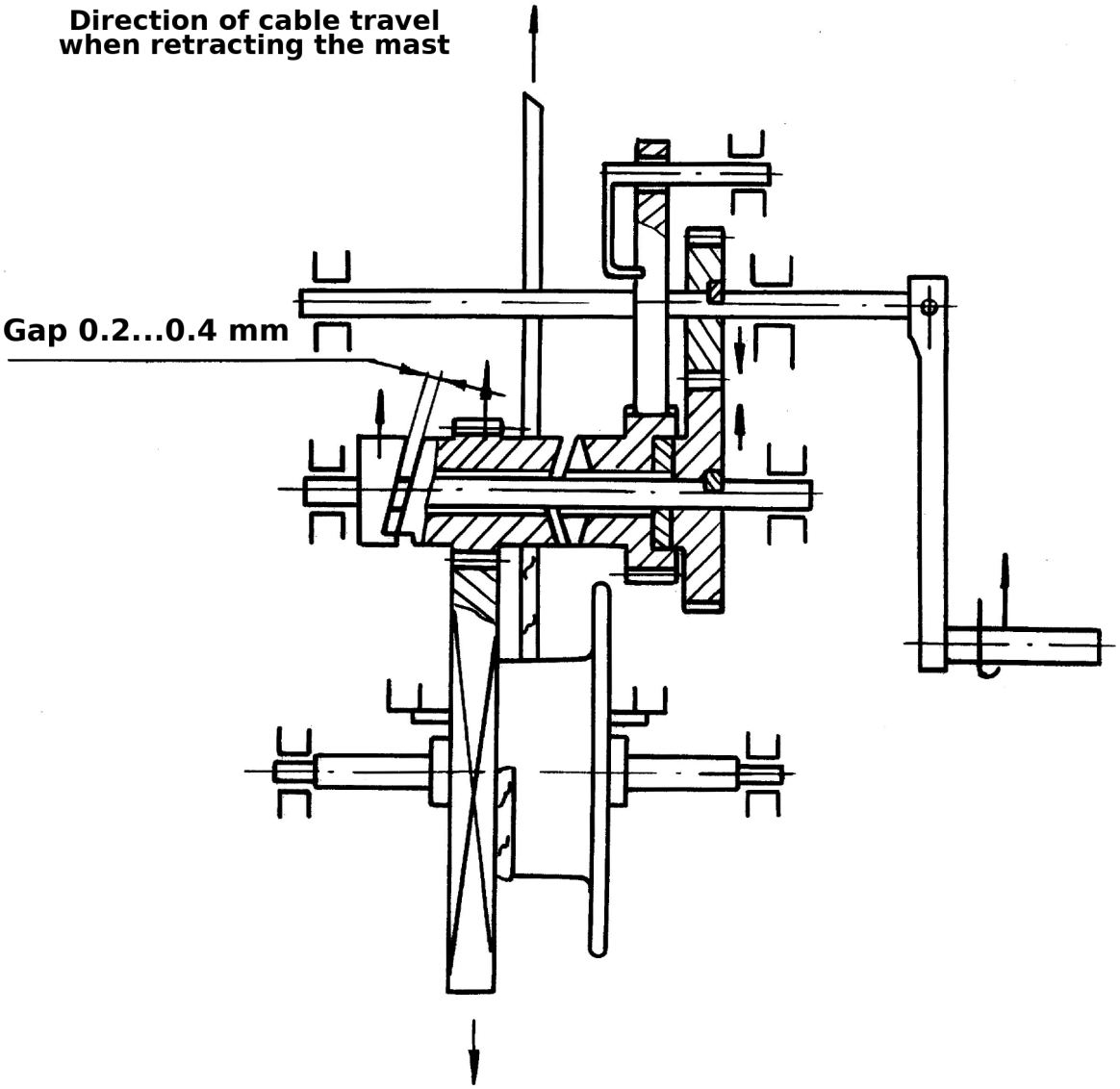


Fig. 13. Winch mechanism diagram
(arrows show the direction of rotation during retraction of the mast)

4. Marking and Sealing

The winch housing bears a stamp indicating the mast's serial number. Masts accepted by the manufacturer's quality control department (OTK) are sealed.

OPERATING INSTRUCTIONS

5. General Instructions

When operating the mast, follow this technical description, the operating instructions, and other governing documents.

Before operating the telescopic mast, check the following: that there are no dents in the mast sections; that the fasteners securing the winch to the fixed section of the mast are tightly torqued; and the quality of the attachment of the clamps on the telescopic mast tubes (the clamps must not be able to rotate relative to the tubes of the telescopic mast).

6. Safety Precautions

When studying the safety instructions and the rules for operating the telescopic mast, be guided by this description and the operating instructions.

Only personnel who have studied this technical description and operating instructions and have passed a safety test are permitted to perform maintenance on the telescopic mast.

When operating the telescopic mast, and also when performing maintenance and scheduled work, it is prohibited to deviate from the order and sequence of deployment (retraction) set out in the operating instructions.

Personnel must remember that careless or unskilled handling of the equipment, and failure to follow the operating instructions and safety precautions, can lead to accidents and cause the equipment to fail.

During deployment and retraction, persons not directly involved in the work are prohibited from being closer than 20 m to the mast shaft.

For repair and scheduled maintenance work, only tools that are in good working order and appropriate for the task may be used.

7. Requirements for the Deployment Site and Pad

Deploying the telescopic mast requires a level, unsloped pad measuring 15×15 m.

The telescopic mast may be operated in areas where the wind speed does not exceed 30 m/s.

8. Marking Out the Pad

The center of the pad is the mast's base plate, if the mast is to be installed on the ground, or the vertical axis of the mast, if it is mounted on a vehicle.

When marking out the deployment pad, the first point must be marked in the direction of one of the mast's lugs; therefore, when the mast is installed on the ground, the points for driving in the stakes must be determined at the same time as the mast is erected to the vertical position.

Determine the stake points in the following sequence:

take the first-tier guy wires and unwind them to their full length (the turnbuckle studs must be screwed out as far as they will go);

place one of the hooks at the center of the base plate (if the mast is mounted on a vehicle, at the point on the ground corresponding to the center of the mast);

stretch the guy wire out to its full length and place the second hook on the ground;

mark the first stake point.

Find the other two points in the same way. All three points are positioned at 120° to one another. Drive the stakes into the marked points with a sledgehammer.

9. Deploying the Telescopic Mast

9.1. Erecting the Telescopic Mast to the Vertical Position

Erect the telescopic mast to the vertical position; the four-sided projection on the mast base must engage the four-sided hole in the plate.

Hook the upper hooks of the first-tier guy wires onto the lower lugs of the telescopic mast (the upper hooks are located on the part of the guy wires without turnbuckles). The turnbuckle studs must be screwed out as far as they will go. Hook the lower hooks onto one of the chain links, providing preliminary tension on the guy wires, then fully tension the guy wires using the turnbuckles. The mast must stand vertically. If the mast is mounted on a vehicle, the first-tier guy wires are not used.

Unwind the guy wires of the remaining tiers to their full length toward the stakes, and hook the upper hooks of the second-tier guy wires onto the upper lugs of the mast; hook the upper hooks of the third-tier guy wires onto the bracket used to fasten the antenna device to the top of the mast.

9.2. Deploying the Telescopic Mast

Deploying the mast without the antenna device installed is strictly prohibited.

Deploy the telescopic mast using a crew of five. Three crew members hold the lower ends of the guy wires (excluding the first tier), and one crew member stands at the winch handle. The fifth (the team leader) stands to one side, at the windward guy wire. On the leader's command “begin raising the mast,” the crew member at the handle begins turning it clockwise, while the crew members at the stakes keep the mast vertical as it deploys by taking up slack in the guy wires and smoothly paying them out.

As the mast deploys, the team leader must watch closely to ensure the mast does not lean, and must promptly indicate the direction of any lean, while the crew members at the stakes correct the lean by tensioning the corresponding guy wires.

Once deployment of the mast is complete, hook the lower hooks of the guy wires onto the chain links at the stakes, providing preliminary tension. Apply final tension using the turnbuckles.

After this, the telescopic mast must stand exactly vertical, with no sagging.

For extended operation of the telescopic mast in the deployed position, it is recommended that, after it is fully deployed, the winch handle be turned back one or two turns in the retracting direction, and the mast then secured in that position using the guy wires.

10. Retracting the Telescopic Mast

Before retracting the mast after extended operation in the deployed state, or in icy conditions after the guy wires have been slackened, turn the winch handle one or two turns clockwise.

Retract the telescopic mast in the reverse order of deployment. Exercise the same caution during retraction as during deployment.

During retraction, turn the winch handle counterclockwise at a rate that allows lowering to be completed in 1.5–2 minutes. Retraction is carried out by a crew of five, with the same roles for its members as during deployment. During retraction, the crew members at the guy wires pull evenly, taking up slack and increasing or decreasing tension on the leader's

command, thereby keeping the mast vertical. During lowering, the leader must also ensure that all sections descend evenly. If any section becomes stuck, stop turning the winch handle and eliminate the cause of the sticking.

After retracting the mast, wind the guy wires onto the turnbuckles, pull out the stakes securing the guy wires, fold the winch handle, and fasten it to the winch lever with the strap.

11. Typical Faults and Remedies

The quickest and most reliable way to find a fault is a step-by-step approach: first identify the faulty assembly, then the defective part within it. Typical faults and their remedies are given in Table 2.

Table 2

Description of fault	Probable cause	Remedy	Remarks
1. Increased force required at the winch handle when deploying the mast.	Outer surfaces of all or some sections are dry (no lubricant). Not all pulleys in the clamps rotate freely. One or more clamps are misaligned (all clamps must be secured in the same plane).	Wipe the mast section surfaces clean and lubricate them. Lubricate the pulley axles. Loosen the clamp fastening, rotate it until it aligns with the plane of the other clamps, and tighten the fastening, making sure the pulley can still rotate.	
2. Increased force required at the winch handle when retracting the telescopic mast.	The lubricant in the friction section of the brake unit has dried out. The brake unit does not engage.	Remove the winch cover, take the brake unit out of the winch, and lubricate the rubbing surfaces of the ratchet wheel, the bronze conical bushing, and the gear. Disassemble the brake unit, wash all parts in gasoline, wipe them dry, re-lubricate all parts, and reassemble.	
3. Sounds typical of unlubricated rubbing surfaces can be heard.	No lubricant in the pulleys.	Lubricate the axles through the grease nipples.	
4. The pulley axle mounted in the telescopic mast's clamp rotates (turns when it should not).	Burrs have formed in the pulley bore and on the surface of the axle.	Remove the axle with the pulley from the clamp, remove the burrs, wash with gasoline, and lubricate.	
5. Movable sections stick when retracting the mast.	Sections are misaligned.	Use the guy wires to set the mast exactly vertical.	

12. Maintenance

The technical condition of the telescopic mast is checked in order to identify and remedy defects. The main defects are: loosening of threaded connections; contamination of parts; damage to paint coatings; and lack of lubrication.

After the defects have been corrected, check the mast's operability by deploying (retracting) it to its full height in accordance with these instructions.

12.1. Types and Frequency of Maintenance

Preventive work is carried out to keep the telescopic mast in working order under various operating conditions and during storage, and comprises the following types of maintenance: daily maintenance (DM); seasonal maintenance (SM).

During daily maintenance, check that the mast is vertical and that the guy wires are properly tensioned. Without retracting the mast (if it is deployed), clean dirt, snow, and ice from the accessible parts of the mast.

Seasonal maintenance is carried out regardless of how intensively the telescopic mast has been used or for how long. Repair personnel of the unit, or a scheduled-maintenance team where one is provided for in the establishment, may be assigned to carry out seasonal

maintenance.

Carry out seasonal preventive work in the following sequence: retract the mast; place the mast on a wooden platform; clean all outer surfaces of the mast's components of dust and dirt (snow, ice).

Check the serviceability and condition of the mast's components.

During the inspection, check: that the section clamps are securely fastened and cannot rotate around the band rings; that the pulleys on their axles in the clamps rotate freely without binding; that the cable thimbles are fastened in the clamps (they must not be over-tightened); and that the water drain hole in the base of the first section is clear — if it is clogged, clean it out.

Check the condition of the lubricant on all parts of the mast.

12.2. Maintenance Procedure

To ensure the telescopic mast operates normally, it is recommended that the lubricant on all rubbing surfaces of the mast be renewed after every 60–70 deployments (but at least once every six months). Remove the old lubricant using rags soaked in gasoline. Apply the new lubricant to clean, dry surfaces with a brush. The lubricant must be free of contamination and must evenly cover the rubbing surfaces. Lubricate the pulley and winch axles using a grease gun.

Mast lubrication procedure:

place the mast on two supports;

unscrew the mounting screws of the base of the first (fixed) section, remove the base, and renew the lubricant on the pulley in the base of the second section;

on the clamp of the first mast section, release the winch cable thimble, insert a round rod into it, and, pulling the cable by the rod, extend the second section as far as it will go (keeping the mast axis straight and free of misalignment);

place a support under the ends of the extended sections.

Remove the old lubricant from the surface of the extended section and from the cable. Inspect the cable; if more than 20% of the wires are broken, replace it with a spare. Apply new lubricant to the cleaned surfaces;

unscrew the bolts securing the winch cover;

remove the cover and clean the old lubricant off the winch parts.

Using an automotive grease gun, inject lubricant into all the winch's lubrication points. Apply lubricant to all the winch parts.

Fit the cover onto the winch and secure it with the bolts;

retract the movable sections into the fixed section, keeping the cable taut to prevent it from twisting inside the section and jamming;

make sure the cable is correctly positioned inside the first section (it must lie on the pulley in the base of the second section); then refit the base of the first section and secure it with the screws.

Secure the winch cable thimble to the clamp of the first section with the bolt;

release the two thimbles of the third section on the clamp of the first section, insert a round rod into them, and by pulling on it extend the third section as far as it will go (place a support under the ends of the sections extended together with the third section);

remove the old lubricant from the surface of the extended section and apply new lubricant.

Before lubricating the cable, inspect its condition and check that the wires are intact. Lubricate the pulley axle through the hole in it, using a grease gun. Retract the third section into the second, keeping the cable taut.

Secure the cable thimble of the third section to the clamp of the first section with the bolt;

lubricate all the remaining movable sections in turn.

The lubrication procedure for these sections is the same as for the third section. Note that the cable thimbles of the fourth section must be secured to the clamp of the second section, the thimbles of the fifth section to the clamp of the third section, and so on;

After securing the cable thimbles of the top section, inspect the mast and make sure it is correctly assembled. Carry out a test deployment of the telescopic mast to its full height.

During the work, use the following tools: 12 mm wrench; 14 mm wrench; 17 mm wrench; screwdriver; brush.

Use TsIATIM-201 grease for the rubbing surfaces. Substitute material: LITA.

After every 60–70 deployments of the telescopic mast, carry out the lubrication work as described above.

12.3. Maintenance with the Telescopic Mast Deployed

With the telescopic mast deployed, carry out the daily preventive maintenance work.

13. Preservation and De-preservation

13.1. Preserving the Telescopic Mast

Preservation is the set of measures for temporarily protecting the telescopic mast from environmental effects during storage. Factory preservation is designed to allow the telescopic mast to be stored for two years. Once this period has elapsed, the telescopic mast must be re-preserved.

Only a technically sound telescopic mast that has been checked for operability may be preserved.

Preserve the mast in the following sequence:

retract the telescopic mast;

thoroughly clean all outer surfaces of the mast of dirt and dust;

coat the rubbing parts with TsIATIM-201 grease (substitute material: LITA);

coat all unpainted metal outer surfaces with a thin layer of PVK grease per GOST 19537-74.

Fit a cover over the top of the mast.

For storage, telescopic masts are placed on racks.

Record the preservation of the telescopic mast in the relevant section of the mast's log/data sheet (passport).

Carry out re-preservation after one year of storage.

13.2. De-preserving the Telescopic Mast

When preparing the telescopic mast for operation, it must be de-preserved. De-preservation is carried out to bring the telescopic mast into working condition, as follows:

wipe all metal parts that were coated with grease during preservation using a rag soaked in gasoline;

check that there is no corrosion on the metal surfaces of the telescopic mast;

check that there are no dents or other mechanical damage.

Check for the presence of TsIATIM-201 grease (substitute material: LITA) on the rubbing surfaces; add or replace the grease as necessary.

14. Storage Rules

14.1. General Instructions

Properly organized storage of the telescopic mast in warehouses and depots, together with high-quality maintenance at the established intervals, are the primary conditions for keeping the telescopic mast in a constant state of technical readiness throughout the storage period.

Only telescopic masts that are in good working order are placed into storage.

When storing telescopic masts, follow the instructions given in this section.

14.2. Storage in Warehouses and Depots

Telescopic masts received at a warehouse for storage of less than six months are not removed from their transport packaging and are stored packed, in a covered building.

Telescopic masts received at a warehouse for long-term storage (longer than six months) may be removed from their transport packaging and placed on racks in a covered building.

The ambient temperature during storage of telescopic masts may range from minus 45°C to plus 50°C.

During long-term storage, telescopic masts are subject to the following processes, which adversely affect their preservation: corrosion of metal parts; contamination; and aging of the lubricant.

Preservation of the mast's condition during long-term storage is achieved by preserving the telescopic mast in accordance with subsection 13.1.

When a telescopic mast is stored in conditions of high humidity, the storage building must be continuously or periodically ventilated, and the preserved telescopic mast in storage must be inspected periodically, once every two months.

15. Transportation

The masts may be transported by any mode of transport.

If the masts are not secured to a vehicle chassis during transport, they must be transported in wooden crates that protect the masts from mechanical damage.

It is prohibited to position the packages without observing the warning markings on them.

Total numbered pages in this technical description and operating instructions: **19** (original Russian-language document)

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